

MOD AUDIO · UTILITY / SIGNAL-ROUTING PLUGIN

PORTAL

SINK + SOURCE

Teleport half your signal chain to another CPU core.

SIGNAL PATH STATUS

SINK	SOURCE	ADDED LATENCY
CONNECTED	CONNECTED	2.6 MS

Plugin by falkTX · github.com/falkTX/portal-lv2

USER MANUAL · SEPTEMBER 2026 EDITION

ABOUT THE PLUGIN

What Portal Does

Think of it like a relay race. Normally, every plugin in your pedalboard runs one after another, on a single runner, on a single CPU core — each one has to finish before the next can start. If one plugin in that line is heavy, everyone behind it has to wait, and the whole chain can stumble (an “xrun,” heard as clicks or dropouts).

Portal hands the baton to a second runner. It comes as two plugins that always work as a pair:

Portal Sink (the entry point, shown in yellow) takes the audio in, and **Portal Source** (the exit, shown in blue) hands it back out a moment later — but now running on its own, separate CPU core. The two halves of your pedalboard that used to compete for one core's attention can now run side by side.

The toll for this trick is exactly one audio cycle of latency — about 2.6 ms at the standard 128-sample buffer size — coming out of the Source. For fast, percussive material right at the split point this can occasionally be felt, but for most chains, and especially anything feeding a reverb, delay, or ambient texture, it's inaudible.

Good to know: Portal only works if both halves are on the pedalboard — a lone Sink or a lone Source does nothing. You can only load one Sink and one Source per pedalboard; it splits your chain into two, not into many.

GETTING STARTED

Quick Start

1. Find the busiest part of your pedalboard — usually a cluster of amp sims, drives, or pitch effects, or a heavy reverb/IR loader further downstream.
2. Add **Portal Sink** and place it right after the section you want to keep on the main core.
3. Add **Portal Source** anywhere else on the board — this is where your audio reappears.
4. Connect the output of your chain into Portal Sink's input.
5. Connect Portal Source's output into the start of the rest of your chain.
6. Play through it. The sound should be identical to before, just spread across two cores — check your CPU meter to see the difference.

THE ONE TRICK TO LEARN

Never wire Portal Sink straight into Portal Source. They're not meant to talk to each other directly — each one should have real plugins on either side. Connecting them directly is an unsupported connection and produces noise instead of audio.

REFERENCE

The Controls

Neither Portal Sink nor Portal Source has any sound-shaping knobs — the entire “control” is where you place them and what you plug into them. Each one exposes just two parameters.

ON SCREEN	UNDERLYING PORT	WHAT IT SHOWS
Enable / Bypass	Enabled	Standard on/off footswitch. Both Sink and Source need to be enabled for the tunnel to work — bypassing either one breaks the connection and the audio behind it goes silent.
Status indicator	Status	A small internal readout (not a control you touch) reflecting whether the plugin currently has a healthy connection to its other half. It's there for diagnostics, not for tweaking.

TECHNIQUE

Where to Place the Portal

Because Portal has no knobs, almost all of the skill in using it well is in *placement*. In practice, most pedalboards split naturally into two zones: things that want to happen before the split, and things that don't mind the extra 2.6 ms.

BEST PLACED BEFORE SINK

- › Pitch shifters
- › Octavers
- › Compressors
- › Overdrives, distortions, fuzz
- › Amp sims and amp profile plugins (NAM, AIDA-X, etc.)

BEST PLACED AFTER SOURCE

- Heavy cabinet IR loaders
- Complex synths and ambient textures
- Reverbs
- Delays

These are starting recommendations, not rules — every pedalboard is different, so treat this as a boilerplate and experiment with shifting the split point left or right from there.

Avoiding phasing: never split your signal so that one copy goes through the Portal and the other

doesn't, and then mix the two back together. Because the Portal adds 2.6 ms, recombining a "Portaled" signal with an "instant" dry one causes comb-filtering — a hollow, swooshing phasing sound. If you need a wet/dry blend, do it before the Portal, then send the fully-mixed signal through as a single wet path.

TECHNIQUE

Recipes

CPU relief before the reverb

Place Portal Sink right after your amp sim / drive section, and Portal Source right before a heavy reverb (Dragonfly Reverb, for example) or a set of cabinet IRs. On a board pushed hard with a modulation effect, two IR loaders and a heavy reverb, this single move has been reported to cut apparent CPU load roughly in half — from around 90% down to 55%.

Dual amp-sim parallel paths

Merge your guitar chain into a single path before the Sink — remember, only one Sink and one Source exist per board — then split into two different amp sims right after the Source, routing each to its own physical output. The result: two different tones from the same guitar, sharing the CPU relief.

Delay-and-reverb overflow valve

On a board that's right on the edge of xruns, place the Portal directly in front of your time-based effects block (delay + reverb). These effects tolerate the extra 2.6 ms better than almost anything else, making this the safest default split point to try first.

REFERENCE PEDALBOARDS

[Before Portal — a CPU-heavy board](#)

[After Portal — the same board, split](#)

[A 4-path \(clean/crunch/dist/high-gain\) board using Portal](#)

TECHNIQUE

Tips

One Sink, one Source. You can't chain multiple Portal pairs to create more than two parallel zones —

one pair per pedalboard is the limit.

Never bridge the gap. Anything that starts before the Sink and needs to mix with something after the Source will phase. Keep the two sides fully separate until the final output.

Order runs Source-then-Sink internally. Because Portal Source has no inputs, the host runs it first in the audio graph — invisible to you, but it's why the two plugins can safely hand off audio without a race condition.

Watch for graphical glitches in Firefox. The Portal's rotating on-screen animation can look odd in Firefox; it's cosmetic only and doesn't affect audio. Chrome-based browsers render it correctly.

TROUBLESHOOTING

Questions & Answers

Q — I added both plugins and now I just get noise. What happened?

Portal Sink and Portal Source are almost certainly wired directly to each other instead of to your other plugins. Connecting them straight together is not a supported signal path — put a real plugin (or your board's input/output) on both sides of each one.

Q — Can I use two Portal pairs to split my board into four sections?

No — only one Sink and one Source can be active per pedalboard. If you need more parallelism than that, you'll need to consolidate sections rather than stack Portals.

Q — Will Portal fix my xruns?

It can help a lot if your xruns come from one core being overloaded while others sit idle — that's exactly the situation Portal is built for. It won't help if a single plugin is simply too heavy for any one core to handle, or if the xruns are unrelated to CPU load.

Q — Is the 2.6 ms latency the same on every device?

The added delay is exactly one audio cycle. At the standard 128-sample buffer on a 48 kHz device that works out to about 2.6 ms; run a larger buffer size and the added latency scales accordingly.

Q — Why does my chain sound hollow / swooshy after adding Portal?

You've likely mixed a Portal-routed (delayed) signal with a non-Portal (instant) copy of the same source somewhere downstream. Move your wet/dry blending to before the Portal instead.

Q — Does bypassing Portal Sink or Source mute my signal?

Yes — both halves need to be active for audio to pass through. Bypassing either one breaks the tunnel, so whatever is downstream of the bypassed plugin will go silent.

REFERENCE

Specifications

PLUGINS	Portal Sink, Portal Source (used as a pair)
CATEGORY	Utility
FORMAT	LV2
AUTHOR	falkTX
LICENSE	ISC
SOURCE	github.com/falkTX/portal-lv2
PLATFORMS	MOD Dwarf, MOD Duo X and MOD Duo
ADDED LATENCY	1 audio cycle (~2.6 ms at 128-sample buffer / 48 kHz)
INSTANCE LIMIT	1 Sink + 1 Source per pedalboard
HOST REQUIREMENTS	Multi-threaded per-chain plugin processing, fixed-size audio buffers, no plugin bridging or out-of-process sync
FORUM THREAD	forum.mod.audio/t/introducing-portal/9329

This document was compiled from the plugin's forum introduction, its GitHub repository, and its store listing. It is a community-sourced manual, not an official document from the plugin's author.